

Sparse Autoencoder

$$\tilde{L} = L(x, g(f(x))) + \lambda \Omega(h)$$

↑
sparsity
penalty

$\Omega(h)$ measures the magnitude of the encoding

$$\Omega(h) = \sum_i |h_i|$$

For as learning descriptive features limits overfitting.

Contractive Autoencoders

$$\hat{L} = L(x, g(f(x))) + \lambda \Omega(f(x), x)$$

↑
derivative
penalty

$\Omega(f(x), x)$ magnitude of derivatives of encoding w.r.t. inputs

$$\Omega(h, x) = \sum_i \left\| \frac{\partial h_i}{\partial x} \right\|^2$$

$$\Omega(h, x) = \sum_i \sum_j \left| \left(\frac{\partial h_i}{\partial x_j} \right) \right|$$

Forces the autoencoder to learn encodings that are robust to small changes in input.

Convolutional Autoencoders

Encoder: $h = f(x)$ is a CNN

Decoder: $\hat{x} = g(h)$ is a CNN

Encoder: Convolution + Pooling

Decoder: Transposed convolution
+ Unpooling

Transposed Convolution:

How to do opposite of convolution with stride?

Insert zeros between all elements in all dimensions of the input. Then regular convolution.

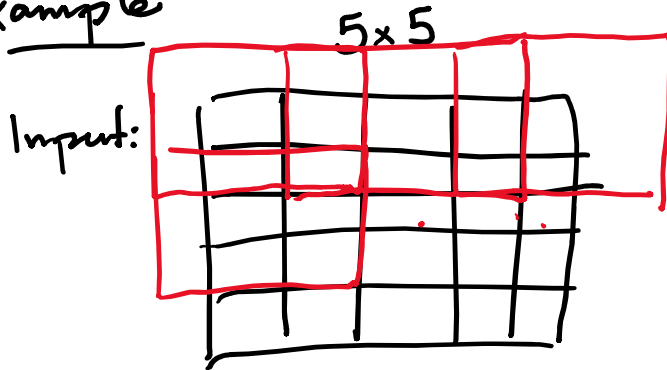
Given a convolution with stride s .

... is a convolution with

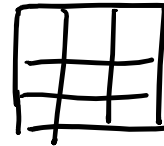
Given a convolution with stride 2.

The opposite convolution is convolution with stride 1 and $s-1$ zeros inserted between elements.

Example

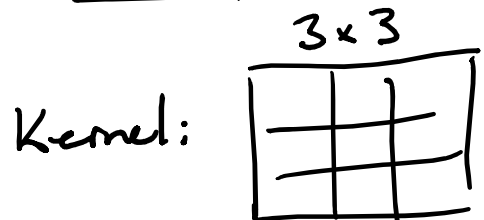
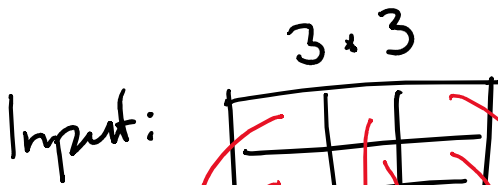
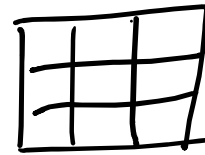


Kernel: 3×3

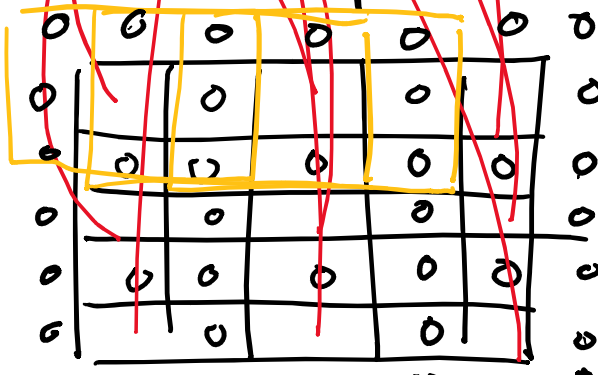


Perform convolution with padding $p=1$ and stride 2.

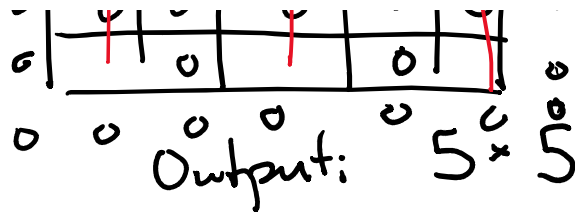
Output: 3×3



We want output 5×5



Padding = 1



Unpooling

Upsampling an image to increase its size.

Imagine we have a pooling operation with kernel size and stride of s .

1. Nearest neighbour:

$$y(i, j) = x(\text{floor}(i/s), \text{floor}(j/s))$$

x :

1	2
3	4

y :

1	1	2	2
1	1	2	2
3	3	4	4
3	3	4	4

2. Bed of nails

$$y(i, j) = \begin{cases} x(i/s, j/s) & \text{if } i/s, j/s \in \mathbb{Z} \\ 0 & \text{otherwise} \end{cases}$$

X:

1	2
3	4

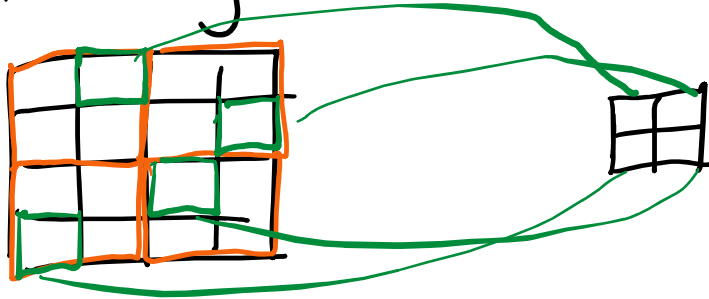
$y:$

1	0	2	0
0	0	0	0
3	0	4	0
0	0	0	0

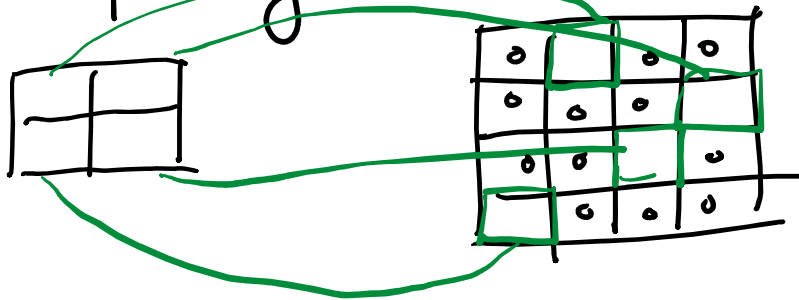
3. Max Unpooling

$$y(i, j) = \begin{cases} x(k, l) & \text{if } i, j \text{ was at a} \\ & \text{maximum in corresponding} \\ & \text{pooling operation} \\ 0 & \text{otherwise} \end{cases}$$

Max Pooling:



Max Unpooling



This requires corresponding max pooling operation in the encoder.

Recurrent Autoencoder

Encoder: $h = f(x)$ is a many-to-one RNN

Decoder: $\hat{x} = g(h)$ is an one-to-many RNN

The hidden state h_T from the RNN encoder is our encoding.